CSE 3544: Practical Programming with C

	ITER, SIKSHA 'O' ANUSANDHAN **(Deemed to be University)**		**LESSON PLAN**
Programme	**B.Tech.**	Academic Year	**2025-2026**
Department	**CSE**	Semester	**5^{th}**
Credit	**4**	Grading Pattern	**1**
Subject Code	**CSE 3544**		
Subject Name	**Practical Programming with C**		
Weekly Course Format	**2L-4P**		
Subject Coordinator (s)	**Pratik Dutta (Coordinator)** **Ishan Ayus (Associate Coordinator)**		
Syllabus	Working with Arrays, Managing Strings, Exploring Functions, Preprocessing and Compilation, Deep Dive into Pointers, File Handling, Implementing Concurrency, Networking and Inter-Process Communication, Sorting and Searching, Working with Graphs, Advanced Data Structures and Algorithms, Creativity with Graphics, General-Purpose Utilities, Dynamic memory allocation, Improving the Performance of Your Code, Low-Level Programming, Applying Security in Coding.		

Text Books(s):

(1) B. M. Harwani, Practical C programming. Birmingham, Mumbai: Packt Publishing, 2020.

Reference Books(s):

(1) J. R. Hanly and E. B. Koffman, Problem solving and program design in C, 8. ed. Global ed. Essex: Pearson Education, 2016.

(2) K. A. Robbins, S. Robbins, and K. A. Robbins, UNIX systems programming, Prentice Hall PTR, 2003.

Course Outcomes	Students will be able to	
	CO1	**write, compile** and **execute** C programs to solve computational problems.
	CO2	**become familiar** with data structures, sorting, searching and string processing.
	CO3	**work** with preprocessor directives and advance data structures using functions & pointers.
	CO4	**understand** processes, threads and concurrent execution of processes/threads.
	CO5	**learn** file IO, buffered IO, inter-process communication and advanced IO.
	CO6	**demonstrate** the usages of C in graphs, graphics and **apply** security in coding

CSE 3544 Practical Programming with C

Sl.No.	Lessons/Topics to be covered	Book Reference (sections)	Mapping with COs	Home Work/ Assignments/ Quizzes
1	**Semester Interaction Class:** Introduce the grading pattern, credit, credit structure, course content, course outcomes (COs), program outcomes (POs) and program specific outcomes (PSOs). Introduce the laboratory session of the course, examination pattern, mark distribution and motivation behind the course.	Textbook(s), Curriculum file	**All COs, POs, & PSOs**	
2	**Introduction to C :** Code vs Program vs Process, Types of Programming Languages, Advantages of C over other languages, and Program compilation stages.	Hanly, pg: 1-27	**CO1**	
3	**LAB 001 :** Introduction to C programming in Linux environment. Introduction to gcc commands options. Study of code, pre-processing, compilation, assembler, linker, and executable code.		**CO1**	**Assignment 01**
4	**LAB 002 :** Introduction to Data Types and Operators in C.		**CO1**	
5	Variable Declaration and Data-types, Working with input-output functions (`printf()` and `scanf()`), Logical and Arithmetic operators.	Hanly, pg: 28-40	**CO1**	
6	Introduction to branch control statements (`if`, `if...else`, `if...else` ladder, Switch statements) and loop control statements (for, while)	Hanly, pg: 129-158, pg: 193-228	**CO1**	
7	**LAB 003 :** Problems on branch control statements		**CO1**	**Assignment 02**
8	**LAB 004 :** Problems on loop control statements		**CO1**	
9	Working with Arrays: Introduction to Array, Initialization of array, Using loops to access and Manipulate Arrays.	Harwani Chapter 01	**CO2**	
10	Inserting an element in an array, Finding the common elements in two arrays, Searching for an item using binary search.		**CO2**	**Quiz 01**
11	**LAB 005 :** Programming on Arrays and Matrices (Finding the unique elements in an array, Merging two sorted arrays into a single array)		**CO2**	**Assignment 03**

Sl.No.	Lessons/Topics to be covered	Book Reference (sections)	Mapping with COs	Home Work/ Assignments/ Quizzes
12	**LAB 006 :** Programming on Arrays and Matrices (Finding whether a matrix is sparse, Multidimensional arrays and their manipulation (Multiplying two matrices))		**CO2**	
13	Introduction to strings, Multidimensional strings. String initialization and manipulation.	Harwani Chapter 02	**CO2**	
14	Managing Strings : Determining whether the string is a palindrome.		**CO2**	
15	**LAB 007 :** Programming on Strings (Displaying the count of each character in a string, Counting vowels and consonants in a sentence.)		**CO2**	**Assignment 04**
16	**LAB 008 :** Programming on Strings (Converting the vowels in a sentence to uppercase, Finding the occurrence of the first repetitive character in a string.)		**CO2**	
17	Introduction to Function, Stack storage, Locality of reference. Introduction to recursion.	Harwani Chapter 03	**CO3**	
18	Preprocessing and Compilation (Performing conditional compilation with directives, Using assertions to ensure a pointer is not pointing to NULL.)		**CO3**	**Quiz 002**
19	**LAB 009 :** Study working of function (Finding whether a number is an Armstrong number, Returning maximum and minimum values in an array, Finding the greatest common divisor using recursion.)		**CO3**	**Assignment 05**
20	**LAB 010 :** Programming on Functions and Preprocessor directive (Converting a binary number into a hexadecimal number, Finding whether a number is a palindrome, Catching errors early with compile-time assertions,)		**CO3**	
21	Introduction to pointers (Reversing a string using pointers, Finding the largest value in an array using pointers.)	Harwani Chapter 05	**CO3**	
22	Introduction to structure and union		**CO3**	
23	**LAB 011 :** Programs on pointers: pointer to arrays, pointer to functions	Harwani Chapter 09	**CO3**	**Assignment 06**

Sl.No.	Lessons/Topics to be covered	Book Reference (sections)	Mapping with COs	Home Work/ Assignments/ Quizzes
24	**LAB 012 :** Programs on structures and unions (Print employees name and gross salary using structures, Display your present address using union, Program on nested structure and union)	Harwani Chapter 09	**CO3**	
25	File Handling : Type of I/O, Functions used in file handling	Harwani Chapter 06	**CO4**	
26	Reading a text file and converting all characters after the period into uppercase		**CO4**	
27	**LAB 013 :** Programming on Files (isplaying the contents of a random file in reverse order)		**CO4**	**Assignment 07**
28	**LAB 014 :** Programming on Files (Counting the number of vowels in a file, Replacing a word in a file with another word, Encrypting a file)		**CO4**	
29	Introduction to Process, Threads and Concurrency. Description of Critical section problem and deadlock.	Harwani Chapter 07	**CO5**	
30	Program to handle tasks with threads (Performing a task with a single thread, Performing multiple tasks with multiple threads)		**CO5**	**Quiz 03**
31	**LAB 015 :** Programming on Process and Threads (Process Creation and Management in C, Perform multiple task using Threads)		**CO5**	**Assignment 08**
32	**LAB 016 :** Programming on Threads (Using mutex to share data between two threads)		**CO5**	
33	Introduction to Networking and Inter-Process Communication	Harwani Chapter 08	**CO5**	
34	Communicating between processes using pipes and FIFO		**CO5**	
35	**LAB 017 :** Program on Communication between processes using pipe and FIFO		**CO5**	**Assignment 09**
36	**LAB 018 :** Programming on inter-thread Communication(Share file content using pipe and FIFO, Create child process to use parent data to perform various arithmetic operations.)		**CO5**	

Sl.No.	Lessons/Topics to be covered	Book Reference (sections)	Mapping with COs	Home Work/ Assignments/ Quizzes
37	Communicating between the client and server using socket programming	Harwani Chapter 08	**CO5**	
38	Communicating between processes using a UDP socket		**CO5**	
39	**LAB 019 :** Programming on Client Server model to Share file content using TCP and UDP		**CO5**	**Assignment 10**
40	**LAB 020 :** Program to share data between two uncorrelated process using shared memory using socket.		**CO5**	
41	General-Purpose Utilities: Dynamic memory allocation	Harwani Chapter 14	**CO6**	
42	Introduction to Storage Class		**CO6**	**Quiz 04**
43	**LAB 021 :** Programming on dynamic memory allocation and Storage Class		**CO6**	**Assignment 11**
44	**LAB 022 :** Programming on dynamic memory allocation (creating linked list and trees with dynamic memory allocation)		**CO6**	
41	Improving the Performance of Your Code(Using the register keyword, Taking input faster, Applying loop unrolling)	Harwani Chapter 15	**CO6**	
42	Understanding low-level programming with bitwise operators.	Harwani Chapter 16	**CO6**	
43	**LAB 023 :** Introduction to buffer overflow vulnerability		**CO6**	**Assignment 12**
44	**LAB 024 :** Program to mitigate buffer Overflow attack and improve the security aspect in C programming		**CO6**	

Program Outcomes (POs)

PO	Description
PO1	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems.
PO2	**Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	**Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO4	**Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8)
PO5	**Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO6	**The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)
PO7	**Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8	**Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	**Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
PO10	**Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	**Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Program Specific Outcomes (PSOs)

PSO	Description
PSO1	**Ability to Understand, Analyze and Develop Programs:** The ability to understand, analyze and develop computer programs in the areas related to business intelligence, web design and networking for efficient design of computer-based systems of varying complexities.
PSO2	**Ability to Apply Standard Practices:** The ability to apply standard practices and strategies in software development using open-ended programming environments to deliver a quality product for business success.

Knowledge and Attitude Profile(WK)

WK	Description
WK1	**Natural and Social Sciences:** A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	**Mathematics and Computing Foundations:** Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3	**Engineering Fundamentals:** A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4	**Engineering Specialist Knowledge:** Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5	**Sustainable Design and Operations:** Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6	**Engineering Practice:** Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK7	**Engineering in Society:** Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8	**Research and Critical Thinking:** Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK	Description
WK9	**Ethics and Inclusion:** Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

COs-POs-PSOs Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	1	1	0	3	0	0	0	0	0	2	0	3
CO2	3	3	1	0	3	0	0	0	0	0	2	0	3
CO3	3	3	1	0	3	0	0	0	0	0	2	0	3
CO4	3	3	1	0	3	0	0	0	0	0	2	0	3
CO5	3	3	2	3	3	0	0	0	0	0	2	0	3
CO6	3	3	2	1	3	0	0	0	0	0	2	0	3